

Year 11 – Higher School Certificate Course

Assessment Task 1

2007



Mathematics Extension 2

Time Allowed: 90 Minutes
(plus 5 minutes reading time)

Instructions to Candidates

1. Attempt all questions.
2. All necessary working must be shown.
3. Start each question on a new page.
4. All diagrams are to be at least $\frac{1}{3}$ page each.

Question 1 – (25 marks) – Start a New Page

Marks

- a) Given that $z_1 = 3 - 2i$ and $z_2 = 4 + 3i$, express each of the following in the form $a + bi$, where a and b are real.

(i) $z_2 - z_1$

1

(ii) $z_1 z_2$

1

(iii) $\frac{z_1}{z_2}$

2

(iv) $(\bar{z}_1)^3$

2

- b) (i) Express $\sqrt{3} - i$ in mod-arg form.

3

- (ii) Hence express $(\sqrt{3} - i)^4$ in the form $a + ib$ (where a, b are real)

2

- c) (i) Express $\sqrt{16 - 30i}$ in Cartesian form (ie $a + ib$ with a, b real).

3

- (ii) Hence, express the roots of $z^2 - (1 - i)z + 7i - 4 = 0$ in the form $x + iy$ (x, y are real).

- d) Draw neat sketches on separate Argand diagrams (of at least $\frac{1}{3}$ page in size) of the locus of a point representing a complex number z if:

(i) $\text{Im}(z) < 1$

(ii) $|z - 1 + 2i| \geq 1$

(iii) $|z + 2| = |z - 3i|$

(iv) $z + \bar{z} > \text{Im}(3z) \cap -\frac{\pi}{4} < \arg z < \frac{\pi}{4}$

Question 2 – (25 marks) – Start a new page

Mark:

- a) (i) Prove that for any complex number z , $z\bar{z} = |z|^2$
- (ii) Prove that for any complex numbers z_1 and z_2 , $\overline{z_1 - z_2} = \bar{z}_1 - \bar{z}_2$
- (iii) Suppose that z_1 , z_2 and z_3 are three complex numbers of modulus 1 such that $z_1 + z_2 + z_3 = 0$

Suppose also that z is a complex number of modulus 3.

Using the results in parts (i) and (ii),

(α) show $|z - z_1|^2 = 10 - (z\bar{z}_1 + \bar{z}z_1)$

(β) show $|z - z_1|^2 + |z - z_2|^2 + |z - z_3|^2 = 30$

- b) (i) Use de Moivre's theorem to solve the equation $z^5 = 1$
- (ii) Show that the points representing the five roots of this equation form the vertices of a regular pentagon when they are plotted on the Argand diagram.
- (iii) Find the area of this pentagon.

c) Given that $z = \cos \theta + i \sin \theta$,

(i) show that $z^n + z^{-n} = 2 \cos n\theta$ using de Moivre's theorem.

(ii) Hence, solve the equation $2z^4 - z^3 + 3z^2 - z + 2 = 0$

(hint: divide throughout by z^2 and use the result $\cos 2\theta = 2\cos^2 \theta - 1$)

Question 3 – (25 marks) – Start a new page

Marks

- a) Sketch the locus of the complex number z if

(i) $\arg(z - 2i) = \arg(z + 3 - i)$

(ii) $\arg\left(\frac{z - 2i}{z + 3 - i}\right) = \frac{\pi}{2}$

- b) (i) If w is a complex cube root of unity, show that the other complex root is w^2 .

(ii) Prove that $1 + w + w^2 = 0$ by using two completely different methods.

(iii) Evaluate $(1 + 2w + 3w^2)(1 + 2w^2 + 3w^4)$

- c) (i) Express the roots of the equation $z^5 + 32 = 0$ in modulus/argument form.

(ii) Hence, show that

$$z^4 - 2z^3 + 4z^2 - 8z + 16 = (z^2 - 4\cos\frac{\pi}{5}z + 4) \times (z^2 - 4\cos\frac{3\pi}{5}z + 4)$$

(iii) By equating coefficients in (ii) above, find the values of:

(α) $\cos\frac{\pi}{5} + \cos\frac{3\pi}{5}$

(β) $\cos\frac{\pi}{5} \cdot \cos\frac{3\pi}{5}$

(iv) Hence, find the exact values of $\cos\frac{\pi}{5}$ and $\cos\frac{3\pi}{5}$ in simplest surd form.

Ex 2

Q1 (i) $z_1 - z_2 = (4-3) + (3-2)i = 1 + i$ — |
 (ii) $z_1 z_2 = 12 + 9i - 8i - 6i^2 = 18 + i$ — |
 (iii) $\frac{z_1}{z_2} = \frac{(3-2i)}{(4+5i)} \times \frac{(4-5i)}{(4-5i)} = \frac{6-17i}{16+9} = \frac{6}{25} - \frac{17}{25}i$ — |

(iv) $(z_1)^3 = (5+12i)(3+2i) - 1 = 15+10i+36i-24 = -9+46i$ — |
 or $z_1^3 = [\sqrt{13}(\cos \theta)]^3 = 13\sqrt{13} [\cos 3\theta + i \sin 3\theta] = 13\sqrt{13} [\cos \theta + i \sin \theta] = -9+46i$ — |

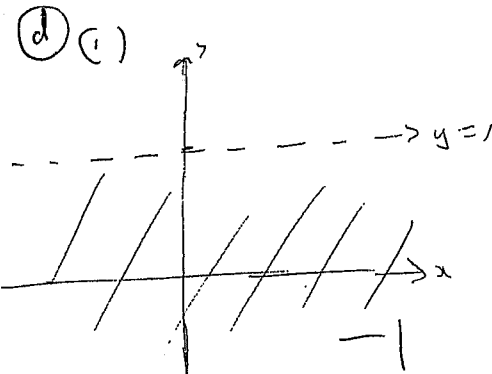
(b) (i) $\sqrt{3}-i = 2(\frac{\sqrt{3}}{2} - \frac{1}{2}i)$ — |
 $\cos \theta = \frac{\sqrt{3}}{2}, \sin \theta = -\frac{1}{2} \Rightarrow \theta = -\frac{\pi}{6}$ — |
 $\therefore \sqrt{3}-i = 2(\cos -\frac{\pi}{6})$ — |
 (ii) $(\sqrt{3}-i)^4 = 2^4 \cos(-\frac{\pi}{6} \times 4) = 16 [\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}] = 16(-\frac{1}{2} + \frac{\sqrt{3}}{2}i) = -8 + 8\sqrt{3}i$ — |

(c) (i) let $\sqrt{16-30i} = a+ib$ — |
 $16-30i = (a^2-b^2) + 2abi$ — |
 Equate real & imag parts: $a^2-b^2=16$ — A
 $2ab=-30$ — B $b = -\frac{15}{a}$

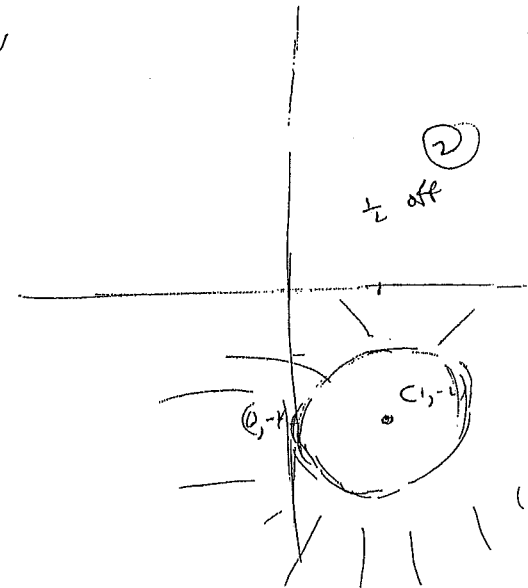
So $a^2 - \frac{225}{a^2} = 16$ — |
 $a^4 - 16a^2 - 225 = 0$ — |
 $(a^2-25)(a^2+9) = 0$ — |
 $a = \pm 5, b = \mp 3$ — |
 $\therefore 5-3i, -5+3i$ — |

(ii) $z = \frac{(1+i) \pm \sqrt{-2i-28i+16}}{2}$ — |
 $= \frac{(1+i) \pm \sqrt{16-30i}}{2}$ — |

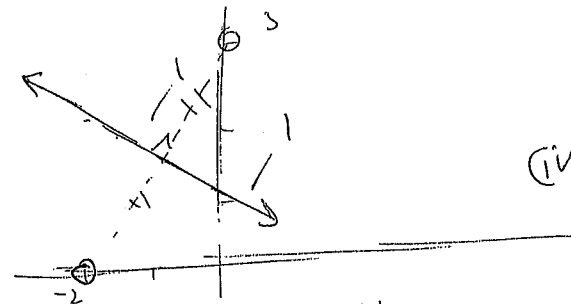
$= 3-2i$ — |, $-2+i$ — |



(ii) $|z - (1-2i)| \geq 1$

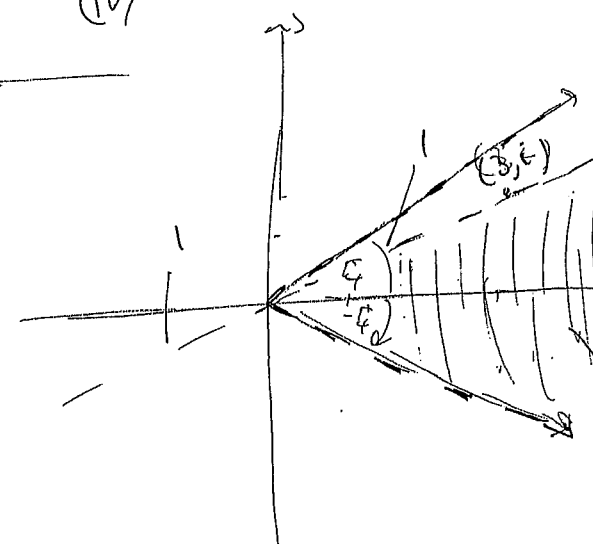


(iii) $|z-2| = |z-3i|$



(iv)

$|x+iy - (-2)| = |2+iy - (2i)|$ — |
 $|x+2 + iy| = |x + (y-2)i|$ — |
 $(x+2)^2 + y^2 = x^2 + (y-2)^2$ — |
 $x^2 + 4x + 4 + y^2 = x^2 + y^2 - 4y + 4$ — |
 $4x + 6y = 5 \Rightarrow y = -\frac{2}{3}x + \frac{5}{6}$ — |
 $2x^2 = 2x$ — |
 $2x > 3y$ — |



Q4
 (1) Let $z = x+iy$
 then $\bar{z} = x-iy$

$\therefore z\bar{z} = (x+iy)(x-iy) \quad \& \quad |z| = \sqrt{x^2+y^2}$
 $= (x^2+y^2) \quad \therefore |z|^2 = x^2+y^2$
 $\therefore z\bar{z} = |z|^2$

(i) $(x_1+iy_1) - (x_2+iy_2)$
 $= (x_1-x_2) + (y_1-y_2)i$
 $\therefore \overline{z_1-z_2} = (x_1-x_2) - (y_1-y_2)i$
 $\underline{L.H.S} = R.H.S.$

(iii) (a) $|z-z_1|^2 = (z-z_1)(\overline{z-z_1})$ for all z
 $= (z-z_1)(\bar{z}-\bar{z}_1)$
 $= z\bar{z} - z\bar{z}_1 - \bar{z}z_1 + z_1\bar{z}_1$
 $= |z|^2 - z\bar{z}_1 - \bar{z}z_1 + |z_1|^2$
 $= 10 - (z\bar{z}_1 + \bar{z}z_1)$

(b) $|z-z_1|^2 + |z-z_2|^2 + |z-z_3|^2$
 $\text{for } z = 10 - (z\bar{z}_1 + \bar{z}z_1) + 10 - (z\bar{z}_2 + \bar{z}z_2) + 10 - (z\bar{z}_3 + \bar{z}z_3)$
 $= 30 - [z(\bar{z}_1 + \bar{z}_2 + \bar{z}_3) + \bar{z}(z_1 + z_2 + z_3)]$

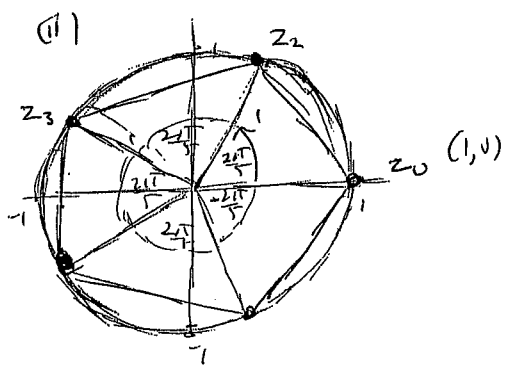
Given $z_1+z_2+z_3=0$
 $\bar{z}_1+\bar{z}_2+\bar{z}_3=0$
 $\therefore (x_1+x_2+x_3) + i(y_1+y_2+y_3) = 0$
 $\therefore (x_1+x_2+x_3) - i(y_1+y_2+y_3) = 0$
 Equate real / im.

$= 30 - (z \times 0 + \bar{z} \times 0)$
 $= 30$

(6) (i) if $\cos \theta = 1 + 0i$
 then $\cos \theta = 1 \quad \& \quad \sin \theta = 0$
 $\therefore \theta = 2k\pi \quad (k=0, 1, 2, 3, 4 \dots \text{repeat})$

$\therefore z^5 = (\cos 2k\pi + i \sin 2k\pi)$
 $z = (\cos \frac{2k\pi}{5} + i \sin \frac{2k\pi}{5})$ by De Moivre's

$n=0, \quad z_0 = 1$
 $n=1, \quad z_1 = \cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5} = \cos \frac{2\pi}{5}$
 $n=2, \quad z_2 = \cos \frac{4\pi}{5} + i \sin \frac{4\pi}{5} = \cos \frac{4\pi}{5} \quad [-\cos \frac{\pi}{5} + i \sin \frac{\pi}{5}]$
 $n=3, \quad z_3 = \cos \frac{6\pi}{5} + i \sin \frac{6\pi}{5} = \cos \frac{6\pi}{5} = \cos(-\frac{4\pi}{5}) = \bar{z}_2$
 $n=4, \quad z_4 = \cos \frac{8\pi}{5} + i \sin \frac{8\pi}{5} = \cos \frac{8\pi}{5} = \cos(-\frac{2\pi}{5}) = \bar{z}_1$



Five points equally spaced at $\frac{2\pi}{5}$ radians on unit circle.

(iii) $A_1 = \frac{1}{2} \times 1 \times 1 \times \sin(\frac{2\pi}{5}) \quad \therefore \text{Total Area}$
 $= 5 \left[\frac{1}{2} \sin(\frac{2\pi}{5}) \right]$
 $= \frac{5}{2} \sin(\frac{2\pi}{5}) \text{ sq units}$

(c) $z^n = (\cos \theta + i \sin \theta)^n$ by De Moivre's
 $= \cos n\theta + i \sin n\theta$

$z^{-n} = \cos(-n\theta) + i \sin(-n\theta)$
 $= \cos n\theta - i \sin n\theta$

Since $\cos(-\theta) = \cos \theta$
 $\sin(-\theta) = -\sin \theta$

$$\therefore z^n + \bar{z}^n = (\cos n\theta + i \sin n\theta) + (\cos n\theta - i \sin n\theta) \\ = 2 \cos n\theta$$

$$(i) \quad 2z^4 - z^3 + 3z^2 - z + 2 = 0$$

$$\div z^2 \\ \therefore 2z^2 - z + 3 - z^{-1} + 2z^{-2} = 0$$

$$\Rightarrow 2(z^2 + z^{-2}) - (z + z^{-1}) + 3 = 0$$

$$4 \cos 2\theta - 2 \cos \theta + 3 = 0$$

$$4(2\cos^2\theta - 1) - 2\cos\theta + 3 = 0$$

$$8\cos^2\theta - 2\cos\theta - 1 = 0$$

quadratic $\cos\theta = \frac{2 \pm \sqrt{4 - 4 \times 8 \times (-1)}}{2 \times 8}$

$$= \frac{2 \pm \sqrt{36}}{16}$$

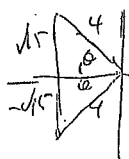
$$= \frac{-4}{16} \quad \text{and} \quad \frac{8}{16}$$

$$= -\frac{1}{4} \quad \frac{1}{2}$$

$$\cos\theta = -\frac{1}{4}$$

In 2nd & 3rd qtr.

$$\theta = \pi - \dots, \pi + \dots$$



$$\therefore z = -\frac{1}{4} \pm \frac{\sqrt{15}}{4}i$$

$$z_1 = -\frac{1}{4}(1 - \sqrt{15}i)$$

$$z_2 = -\frac{1}{4}(1 + \sqrt{15}i)$$

$$\cos\theta = \frac{1}{2}$$

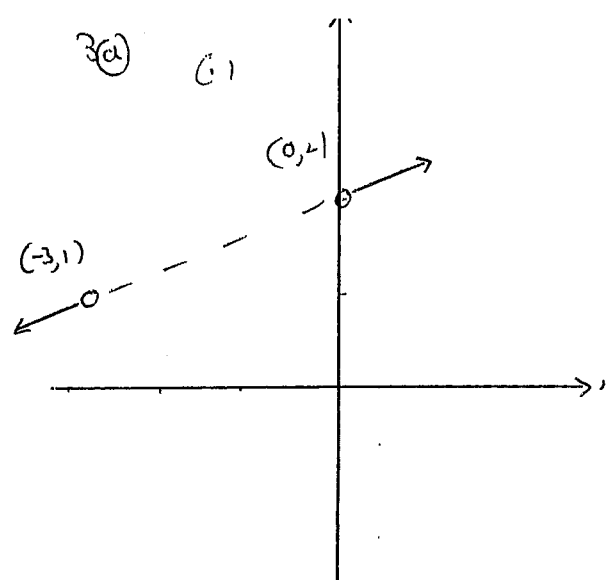
In 1st & 4th qtr.

$$\theta = \frac{\pi}{3}$$

$$z_3 = \frac{1}{2} + \frac{\sqrt{3}}{2}i$$

$$z_4 = \frac{1}{2} - \frac{\sqrt{3}}{2}i$$

3(a) (i)



$$\arg(z - 2i) = \arg(z - (-3 + i))$$

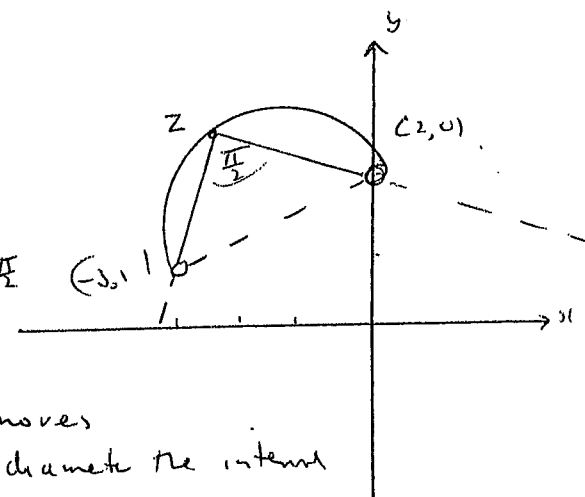
- line through (0, 2) and (-3, 1), excluding interval between points

$$(ii) \quad \arg\left(\frac{z-2i}{z-(-3+i)}\right) = \frac{\pi}{2}$$

$$\text{the } \arg(z-2i) + \arg(z-(-3+i)) = \frac{\pi}{2}$$

$$\Rightarrow \arg(z-2i) = \arg(z-(-3+i)) + \frac{\pi}{2}$$

By ext. $\angle$ of Δ , z moves on semi-circle with diameter the interval joining (0, 2) to (-3, 1)



$$(b) \quad \text{If } \omega^3 = 1$$

$$\text{then } \omega^3 - 1 = 0 ; (\omega - 1)(\omega^2 + \omega + 1) = 0$$

$$\text{Now } \omega^2 + \omega + 1 = 0 \quad \text{use quad. formula} \quad \omega = \frac{-1 \pm \sqrt{1-4}}{2} \\ (\text{use } i^2 = -1) \quad = -\frac{1}{2} \pm \frac{i\sqrt{3}}{2}$$

$$\omega = 1, -\frac{1}{2} - \frac{i\sqrt{3}}{2}, -\frac{1}{2} + \frac{i\sqrt{3}}{2}$$

$$\text{+ } \omega^2 = \left(\frac{1}{4} - \frac{3}{4}\right) - 2i\frac{\sqrt{3}}{4}$$

$$[\text{complex}] = -\frac{1}{2} - i\frac{\sqrt{3}}{2} \rightarrow = \bar{\omega}$$

(ii) Method ① $1 + \omega + \omega^2 = 1 + \left(-\frac{1}{2} + \frac{\sqrt{3}i}{2}\right) + \left(-\frac{1}{2} - \frac{\sqrt{3}i}{2}\right)$
 $= 1 - 1$
 $= 0$ as required

② $1 + \omega + \omega^2$ is a Geometric Series
 $\therefore S_3 = 1 \frac{(\omega^3 - 1)}{\omega - 1}$
 $= \frac{\omega^3 - 1}{\omega - 1}$ since ω is a cube root of unity
 $\omega^3 - 1 = 0$
 $S_3 = 0$

or $1 + \omega + \omega^2 = 0$ as required

$(z-1)(z-\omega)(z-\omega^2) = (z^2 - (1+\omega)z + \omega)(z-\omega^2)$
 $= z^3 - (1+\omega)z^2 + \omega z - z^2\omega^2 + (1+\omega)\omega z - \omega^3$
 $= z^3 - (1+\omega+\omega^2)z^2 + (\omega^2+\omega+1)z + \omega^3$
 $= z^3 - 0z^2 + 0z + 1$
 $= z^3 - 1$

Now equate coefficients $\left\{ \begin{array}{l} 1 + \omega + \omega^2 = 0 \\ \text{on } z^2 \end{array} \right.$

(iii) $(1+2\omega+3\omega^2)(1+2\omega^2+3\omega)$ $\omega^4 = \omega$
 $= (1+2\omega+3\omega^2)(1+2\omega^2+3\omega)$
 $= (1+\omega+\omega^2-1+\omega^2)(2(1+\omega+\omega^2)-1+\omega)$
 $= (\omega^2-1)(\omega-1)$
 $= \omega^3 - \omega^2 - \omega + 1$
 $= 2 - (\omega + \omega^2)$
 $= 3 - (1 + \omega + \omega^2)$
 $= 3 - 0$
 $= 3$

② (i) If $re^{i\theta}$ is a root
then $re^{i\theta} = -32 + 0i$
 $32 \cos \theta = -32$ $\therefore 32 \sin \theta = 0$
 $r = 32$ $\cos \theta = -1$ $\theta = (2k+1)\pi$ $k = 0, 1, 2, 3, 4$

$\therefore z^5 = +32 (\cos (2k+1)\pi + i \sin (2k+1)\pi)$
 $z = +2 (\cos (\frac{(2k+1)\pi}{5}) + i \sin (\frac{(2k+1)\pi}{5}))$

$k=0, z_0 = 2 \cos \frac{\pi}{5} = 2 [\cos \frac{\pi}{5} + i \sin \frac{\pi}{5}]$

$k=1, z_1 = 2 \cos \frac{3\pi}{5} = 2 [\cos \frac{3\pi}{5} + i \sin \frac{3\pi}{5}]$

$k=2, z_2 = 2 \cos \pi = -2$

$k=3, z_3 = 2 \cos \frac{7\pi}{5} = 2 [-\cos \frac{2\pi}{5} - i \sin \frac{2\pi}{5}] = \bar{z}_1 = 2 \cos(-\frac{3\pi}{5})$

$k=4, z_4 = 2 \cos \frac{9\pi}{5} = 2 [\cos \frac{\pi}{5} - i \sin \frac{\pi}{5}] = \bar{z}_0 = 2 \cos(-\frac{\pi}{5})$

(ii) $z^5 + z^5 = (z-z_0)(z-z_1)(z-z_2)(z-z_3)(z-z_4)$
 $= (z-z_0)(z-\bar{z}_0)(z-z_1)(z-\bar{z}_1)(z-z_2)$
 $= [z^2 - (z_0+\bar{z}_0)z + z_0\bar{z}_0][z^2 - (z_1+\bar{z}_1)z + z_1\bar{z}_1](z-z_2)$
 $= [z^2 - 4 \cos \frac{\pi}{5} z + 4][z^2 - 4 \cos \frac{3\pi}{5} z + 4][z - (-2)]$

Now, $(z+\bar{z}) = 2 \operatorname{Re}(z)$ $\therefore z\bar{z} = |z|^2 = 4$

$\therefore (z+2)(z^4 - 2z^3 + 4z^2 - 8z + 16)$
 $= z^5 - 2z^4 + 4z^3 - 8z^2 + 16z + 2z^4 - 4z^3 + 8z^2 - 16z + 32$
 $= z^5 + 32$

Let $(z+2)(z^4 - 2z^3 + 4z^2 - 8z + 16) = (z+2)[z^2 - 4 \cos \frac{\pi}{5} z + 4][z^2 - 4 \cos \frac{3\pi}{5} z + 4]$

$\therefore z^4 - 2z^3 + 4z^2 - 8z + 16 = [z^2 - 4 \cos \frac{\pi}{5} z + 4][z^2 - 4 \cos \frac{3\pi}{5} z + 4]$

(iii) Equate coeff

$$\begin{aligned} \textcircled{x} \quad z^4 - 4z^3 \cos \frac{3\pi}{5} + 4z^2 - 4z^3 \cos \frac{\pi}{5} + 16z^2 \cos \frac{\pi}{5} \cos \frac{3\pi}{5} \\ - 16z \cos \frac{\pi}{5} + 4z^2 - 16z \cos \frac{3\pi}{5} + 16 \\ = z^4 - z^3 (4 \cos \frac{3\pi}{5} + 4 \cos \frac{\pi}{5}) + z^2 (16 \cos \frac{\pi}{5} \cos \frac{3\pi}{5} + 8) \\ - 16z (\cos \frac{\pi}{5} + \cos \frac{3\pi}{5}) + 16 \end{aligned}$$

$$\begin{aligned} \textcircled{x} \text{ on } z^3 \quad -(4 \cos \frac{\pi}{5} + 4 \cos \frac{3\pi}{5}) &= -2 \\ \cos \frac{\pi}{5} + \cos \frac{3\pi}{5} &= \frac{1}{2} \end{aligned}$$

$$\begin{aligned} \textcircled{b} \text{ on } z^2 \quad 16 \cos \frac{\pi}{5} \cos \frac{3\pi}{5} + 8 &= 4 \\ \cos \frac{\pi}{5} \cos \frac{3\pi}{5} &= -\frac{1}{4} \end{aligned}$$

$$\begin{aligned} \text{(iv)} \quad \textcircled{x} \quad \cos \frac{3\pi}{5} &= \frac{1}{2} - \cos \frac{\pi}{5} \implies \cos \frac{\pi}{5} \left(\frac{1}{2} - \cos \frac{\pi}{5} \right) = -\frac{1}{4} \\ \frac{1}{2} \cos \frac{\pi}{5} - \cos^2 \frac{\pi}{5} &= -\frac{1}{4} \\ 4 \cos^2 \left(\frac{\pi}{5} \right) - 2 \cos \frac{\pi}{5} - 1 &= 0 \\ \therefore \cos \frac{\pi}{5} &= \frac{2 \pm \sqrt{4+16}}{8} \\ &= \frac{2 \pm 2\sqrt{5}}{8} \\ &= \frac{1+\sqrt{5}}{4} \quad \& \quad \frac{1-\sqrt{5}}{4} \end{aligned}$$

also

$\cos \frac{3\pi}{5}$ gives same results

$$\therefore \cos \frac{\pi}{5} = \frac{1+\sqrt{5}}{4} > 0$$

$$\cos \frac{3\pi}{5} = \frac{1-\sqrt{5}}{4} < 0$$